Minor Frame Word	Subcom Word	Bit	Unit ID	Telemetry Descriptor	Analog or Digital	Analog Range or Digital States	Remarks
0			MEH	Cosmic Ray Rlectron Data	D		
1			↓	Cosmic Ray Electron Data	D		See B-XIII App B
2				Cosmic Ray Electron Data	D		
3			MEH	Cosmic Ray Rlectron Data	D		
4			SBH	Am Exp. Data	A		See B-XV, App B
5			TYH	VLET PHA Data	D		
6				VLET PHA Data	D		
7				VLET Rates and PHA Status	D		See B-XX, App B
8				HET Rates	D		
9				HET PHA Data	D		
10				HET PHA Data	D		
11			↓	HET PHA Data	D		
12			TYH	Tel 1 Exp Data	D		
13			DFH	Tel 1 Exp Data	D		See B-X, App B
14				Tel 2 Exp Data	D		
15				Tel 2 Exp Data	D		
16				Tel 3 Exp Data	D		
17				Tel 3 Exp Data	D		
18			↓	Exp Data	D		See B-XII, App B
19			HOH	Exp Data	D		See B-XV, App B
20			SBH	Am Exp Data	A		
21			HOH	Exp Data	D		See B-XII, App B
22				Exp Data	D		
23			↓	Exp Data	D		
			HOH				

Minor Frame Word	Subcom Word	Bit	Unit ID	Telemetry Descriptor	Analog or Digital	Analog Range or Digital States	Remarks
24			HOH	Exp Data	D		
25			HOH	Exp Data	D		
26			HOH	Exp Data	D		See B-XII, App B
27			SMH	Triaxial Mag Fld Data	D		
28				Triaxial Mag Fld Data	D		See B-XVII, App B
29				Triaxial Mag Fld Data	D		
30			SMH	Triaxial Mag Fld Data	D		
31			HKH	Cosmic Ray Event Data	D		
32			HKH	Cosmic Ray Event Data	D		See B-XI, App B
33			BAH	Solar Wind Ions Data	D		
34			BAH	Solar Wind Ions Data	D		See B-IX, App B
35			BAH	Solar Wind Ions Data	D		
36			SBH	Am Exp Data	A		See B-XV, App B
37			STH	Exp Data	D		See B-XVIII, App B
38			OGH	Exp Data	D		See B-XIV, App B
39			STH	Exp Data	D		See B-XVIII, App B
40			ANH	X-Ray Experiment Data	D		
41				X-Ray Experiment Data	D		
42				X-Ray Experiment Data	D		
43				X-Ray Experiment Data	D		See B-VIII, App B
44				X-Ray Experiment Data	D		
45				X-Ray Experiment Data	D		
46				X-Ray Experiment Data	D		
47			ANH	X-Ray Experiment Data	D		
48			STH	Exp Data	D		See B-XVIII, App B

Minor Frame Word	Subcom Word	Bit	Unit ID	Telemetry Descriptor	Analog or Digital	Analog Range or Digital States	Remarks
49			BAH	Solar Wind Ions Data	D		See B-IX, App B
50			BAH	Solar Wind Ions Data	D		
51			BAH	Solar Wind Ions Data	D		
52			SBH	Am Exp Data	A		See B-XV, App B
53			BAH	Solar Wind Ions Data	D		
54			BAH	Solar Wind Ions Data	D		See B-IX, App B
55			STH	Exp Data	D		See B-XVIII, App B
56			SBH	Exp Ph/Status Data	A		See B-XV, App B
57			STH	Exp Data	D		See B-XVIII, App B
58	0		BAH	+5 Volt Pwr Monitor	A		
	1		BAH	High Voltage A	A		
	2		BAH	High Voltage B	A		See B-IX, App B
	3		BAH	Sweep A Mon	A		
	4		BAH	Sweep B Mon	A		
	5		ANH	X-Ray Mem Logic Volt	A		
	6		ANH	SST Hskg Data	A		See B-VIII, App B
	7		ANH	SST Hskg Data	A		
	8		HOH	Hskg Data	A		
	9		HOH	Hskg Data	A		
	10		HOH	Hskg Data	A		See B-X, App B
	11		HOH	Hskg Data	A		
	12		HOH	GRB PM Volt Mon	A		
	13		ANH	X-Ray Hskg Data	A		See B-VIII, App B
	14		ANH	X-Ray Hskg Data	A		
	15		CM	Wire Ant Deploy (+)	A		
58	16		BAH	Exp Temp	A		

Minor Frame Word	Subcom Word	Bit	Unit ID	Telemetry Descriptor	Analog or Digital	Analog Range or Digital States	Remarks
58	17		AT	Accel Temp	A		
	18		ANH	X-Ray Mem Temp	A		
	19		ANH	X-Ray Temp	A		
	20		ANH	SST Temp	A		
	21		HKH	Drift Chamb Temp	A		
	22		HKH	Silicon Det Temp	A		
	23		MEH	Elec Temp	A		
	24		MEH	CsI Xtal Temp	A		
	25		HOH	UOM Pkg Temp	A		
	26		HOH	MPI Pkg Temp	A		
	27		SCH	Short Ant. Preamp Temp	A		
	28		SCH	B Ant Preamp Temp	A		
	29		STH	Exp Temp	A		
	30		DFH	Cold Plate Temp	A		
	31		DFH	Conv Temp	A		
	32		HKH	Pwr Mon	A		
	33		HHK	Exp Status	A		
	34		HKH	Hskg Data	A		See B-XI, App B
	35		HKH	Hskg Data	A		
	36		CM	Wire Ant Deploy (-)	A		
	37		MEH	Exp Pwr ON Mon	A		See B-XIII, App B
	38			Spare	A		
	39		SCH	Pwr Mon	A		See B-XVI, App B
58	40		STH	Det Temp	A		See B-XVIII, App B

Minor Frame Word	Subcom Word	Bit	Unit ID	Telemetry Descriptor	Analog or Digital	Analog Range or Digital States	Remarks
58	41		DFH	Prm Pwr Conv Mon	A		
	42		DFH	BU Pwr Conv Mon	A		
	43		DFH	Tel 1 Kskg Data	A		See B-X, App B
	44		DFH	Tel 2 Hskg Data	A		
	45		DFH	Tel 3 Hskg Data			
	46		OGH	+ Volt Mon	A		
	47		OGH	High Volt Mon	A		See B-XIV, App B
	48		SBH	B Pwr Mon	A		
	49		SBH	B * Pwr Mon	A		See B-XV, App B
	50			Spare			
	51		TYH	VLETS Hskg Data	A		
	52		TYH	HETS Pwr Mon	A		See B-XX, App B
	53		TYH	VLETS Pwr Mon	A		
	54		AM	FSS-A Coarse (V01)	A		
	55			FSS-A Coarse (V02)	A		
	56			FSS-A Coarse (V03)	A		
	57			FSS-A Fine (VS)	A		
	58		AM	FSS-A Fine (VC)	A		
	59		-	Spare	A		
	60		SMH	Hskg Data	A		
	61		SMH	SA A or X Fld Data	A		
	62		SMH	SA B or Y Fld Data	A		See B-XVII, App B
58	63		SMH	SA C or Z Fld Data	A		

Minor Frame Word	Subcom Word	Bit	Unit ID	Telemetry Descriptor	Analog or Digital	Analog Range or Digital States	Remarks
59	0		AM	PAS Attitude	D		
	1			PAS Attitude			
	2			PAS Attitude			
	3			PAS Attitude			
	4			PAS Attitude			
	5			PAS Attitude			See B-I, App B
	6			PAS Attitude			
	7			PAS Attitude			
	8			PAS Attitude			
	9		AM	PAS Attitude			
	10		AT	HPS Elec A CMD Status	D		
	11			HPS Elec A CMD Status			
	12			HPS Elec A CMD Status			
	13			HPS Elec A CMD Status			
	14			HPS Elec A CMD Status			
	15			HPS Elec B CMD Status			See B-II, App B
	16			HPS Elec B CMD Status			
	17			HPS Elec B CMD Status			
	18			HPS Elec B CMD Status			
	19		AT	HPS Elec B CMD Status			
	20		CM-01	CMD Dcdr B CMD Ctr	D		
	21		CM-01	CMD Dcdr A CMD Ctr	D		
	22		CM-03	Wire Ant CMD Flags			See B-III, App B
59	23		CM-03	Wire Ant Deploy Flags			

Minor Frame Word	Subcom Word	Bit	Unit ID	Telemetry Descriptor	Analog or Digital	Analog Range or Digital States	Remarks
59	24		BAH	EED No. 1 Arm Status	D		See B-IX, App B
	25		BAH	EED No. 2 Arm Status	D		See B-IX, App B
	26			Spare			
	27		SBH	CMD State/Status	D		See B-XV, App B
	28		SBH	CMD State/Status	D		
	29		SCH	Inst Status	D		See B-XVI, App B
	30		OGH	Exp Status Data	D		See B-XIV, App B
	31		OGH	Exp Status Data	D		
	32		TL	Sun Time (8 MSB)	D		
	33			Sun Time (8 LSB)	D		
	34			Sun Period (8 MSB)	D		
	35			Sun Period (8 LSB)	D		
	36			Mag Time (8 MSB)	D		
	37			Mag Time (8 LSB)	D		See B-VII, App B
	38			Mag Period (8 MSB)	D		
	39			Mag Period (8 LSB)	D		
	40			SSC Cmd Reg Status R. O	D		
	41			SSC Cmd Reg Status R. O.	D		
	42		TL	SSC Cmd Reg Status R. O.	D		
	43		TYH	HET Cmd Status	D		
	44			HET Cmd Status	D		See B-XX, App B
	45			HET Cmd Status	D		
59	46		TYH	HET Cmd Status	D		

Minor Frame Word	Subcom Word	Bit	Unit ID	Telemetry Descriptor	Analog or Digital	Analog Range or Digital States	Remarks
59	47			Spare			
	48		PW	Configuration Status	D		See B-IV, App B
	49		SD	Configuration Status	D		See B-V, App B
	50		SD	Configuration Status	D		
	51		AT	Accel Pulse Ctr	D		See B-11, App B
	52		CM	CMD Dcdr B Cmd Ctr	D		See B-III, App B
	53		CM	Cmd Dcdr A CMD Ctr	D		
	54		AT	HPS Htrs Cmd Status	D		See B-11, App B
	55		AT	HPS Htrs Cmd Status	D		
	56			Spare	D		
	57			Spare	D		
	58			Spare	D		
	59		TL	Data System ID	D		See B-VII, App B
	60			Spare	D		
	61		MEH	Inst Status	D		See B-XIII, App B
	62		MEH	Inst Status	D		
59	63		SMH	Inst Status	D		See B-XVII, App B
60	0	1,0	TL	Word 61, 62, 63 Subcom ID	D	Bit 0 = 0 bit 1 = 0	
	1	1,0		Word 61, 62, 63 Subcom ID	D	Bit 0 = 1 bit 1 = 0	
	2	1,0		Work 61, 62, 63 Subcom ID	D	Bit 0 = 0 bit 1 = 1	
60	3	1,0		Word 61, 62, 63 Subcom ID	D	Bit 0 = 1 bit 1 = 1	
61	0	7-0		S/C Clock	D	8 MSB's	
	1	7-0		Variable Mem R/O	D	8 bits	
61	2	7-0	TL	Status Reg	D		See VIII, of Text

Minor Frame Word	Subcom Word	Bit	Unit ID	Telemetry Descriptor	Analog or Digital	Analog Range or Digital States	Remarks
61	3	7-0	TL	Variable Mem R/O	D	8 bits	
62	0	7-0		S/C Clock	D	Bits 9-16	
62	1	7-0		Indirect Add Reg	D	8 bits	
62	2	7-0		Status Reg	D		See VIII of Text
62	3	7-0		CMD Reg	D	Bits 1-8	
63	0	7-0		S/C Clock	D	8 LSB's	
	1	7-0		Execute Add R/O	D	8 bits	
	2	7-0		Status Reg	D		See VIII of Text
63	3	7-0	TL	CMD Reg	D	Bits 9-16	
64			STH	Exp Data	D		See B-XVIII, App B
65			BAH	Solar Wind Ions Data	D		
66			BAH	Solar Wind Ions Data	D		See B-IX, App B
67			BAH	Solar Wind Ions Data	D		
68			SBH	Am Exp Data	A		See B-XV, App B
69			STH	Exp Data	D		See B-XVIII, App B
70			SMH	Triaxial Mag Fld Data	D		
71				Triaxial Mag Fld Data	D		See B-XVII, App B
72				Triaxial Mag Fld Data	D		
73			SMH	Triaxial Mag Fld Data	D		
74			ANH	X-Ray Exp Data	D		
75				X-Ray Exp Data	D		
76				X-Ray Exp Data	D		
77				X-Ray Exp Data	D		See B-VIII, App B
78				X-Ray Exp Data	D		
79			ANH	X-Ray Exp Data	D		

Minor Frame Word	Subcom Word	Bit	Unit ID	Telemetry Descriptor	Analog or Digital	Analog Range or Digital States	Remarks
80			STH	Exp Data	D		See B-XVIII, App B
81			BAH	Solar Wind Ions Data	D		
82			BAH	Solar Wind Ions Data	D		See B-IX, App B
83			BAH	Solar Wind Ions Data	D		
84			SBH	Am Exp Data	A		See B-XV, App B
85			SCH	16 Chan f_{16} (100 KHz) Data	A		
86				16 Chan f_{15} (56 KHz) Data	A		
87				16 Chan f_{14} (31 KHz) Data	A		
88				16 Chan f_{13} (17 KHz) Data	A		
89				16 Chan f_{12} (10 KHz) Data	A		
90				16 Chan f_{11} (5.6 KHz) Data	A		
91				16 Chan f_{10} (3.1 KHz) Data	A		See B-XVI, App B
92				16 Chan f_9 (1.7 KHz) Data	A		
93				16 Chan f_8 (1 KHz) Data	A		
94				16 Chan f_7 (560 Hz) Data	A		
95				16 Chan f_6 (310 Hz) Data	A		
96				16 Chan f_5 (170 Hz) Data	A		
97				16 Chan f_4 (100 Hz) Data	A		
98				16 Chan f_3 (56 Hz) Data	A		
99			SCH	16 Chan f_2 (31 Hz) Data	A		
100			SBH	Am Exp Date	A		See B-XV, App B
101			SCH	16 Chan f_1 (17 Hz) Data	A		
102			SCH	8 Chan Data	A		See B-XVI, App B
103			BAH	Solar Wind Ions Data	D		
104			BAH	Solar Wind Electrons Data	D		See B-IX, App B

Minor Frame Word	Subcom Word	Bit	Unit ID	Telemetry Descriptor	Analog or Digital	Analog Range or Digital States	Remarks
105			BAH	Solar Wind Electrons Data	D		
106				Solar Wind Electrons Data	D		
107				Solar Wind Electrons Data	D		
108				Solar Wind Electrons Data	D		
109				Solar Wind Electrons Data	D		See B-IX, App B
110				Solar Wind Electrons Data	D		
111			BAH	Solar Wind Electrons Data	D		
112			SMH	Triaxial Mag Fld Data	D		
113				Triaxial Mag Fld Data	D		
114				Triaxial Mag Fld Data	D		See B-XVII, App B
115			SMH	Triaxial Mag Fld Data	D		
116			SBH	Am Exp Data	A		See B-XV, App B
117			ANH	SST Exp Data	D		
118				SST Exp Data	D		
119				SST Exp Data	D		See B-VIII, App B
120				SST Exp Data	D		
121			ANH	SST Exp Data	D		
122	0		AM	PAS Pwr Mon	A		
	1		CM	Cmd Dcdr A -10V Mon	A		
	2		CM	Cmd Dcdr B -10V Mon	A		
	3		AT	Accel Pwr Mon	A		
	4		AT	HPS-A TC	A		
	5		AT	HPS-B TC	A		
	6		AM	FSS-A Pwr Mon	A		
122	7		AM	FSS-A Bias Volt	A		

Minor Frame Word	Subcom Word	Bit	Unit ID	Telemetry Descriptor	Analog or Digital	Analog Range or Digital States	Remarks
122	8		AM	FSS-B Bias Volt	A		
	9		CM	Cmd Dcdr A +10V Mon	A		
	10		CM	Cmd Dcdr B +10V Mon	A		
	11		AM	FSS-B Pwr Mon	A		
	12		HOH	Low Volt Conv	A		See B-XII, App B
	13		HOH	Stepped Defl Volt (UL)	A		
	14		AT	HPS-A Tk Press	A		
	15		AT	HPS-B Tk Press	A		
	16		TL	DMU Temp	A		
	17		TYH	HETS Temp	A		
	18		SBH	θ B or θB* Temp	A		See B-XV, App B
	19		TYH	VLETS Temp	A		
	20		AM	FSS Elec Temp	A		
	21		AM	PAS Elec Temp	A		
	22		AM	PAS Sensor Temp	A		
	23		CM	Cmd Dcdr A Temp	A		
	24		PW	Btry Temp	A		
	25		PW	Pwr Module Temp	A		
	26		PW	MAM Temp	A		
	27		PW	Solar Array Temps	A		See B-IV, App B
	28		SD	Xpndr A WB Fil Temp	A		
	29		SD	Xpndr B WB Fil Temp	A		
	30		SD	Xpndr A RF Tstr Temp	A		
	31		SD	Xpndr B RF Tstr Temp	A		
122	32		TL	+3.2V Cal	A		

Minor Frame Word	Subcom Word	Bit	Unit ID	Telemetry Descriptor	Analog or Digital	Analog Range or Digital States	Remarks
122	33		TL	+0.1V Cal	A		
122	34		TL	+5.0V Cal	A		
122	35		TL	+2.5V Cal	A		
122	36		TL	Radiation Mon	A		
122	37		TL	A/D Ramp	A		
122	38		TL	2.5V @ 2 M	A		

Minor Frame Word	Subcom Word	Bit	Unit ID	Telemetry Descriptor	Analog or Digital	Analog Range or Digital States	Remarks
122							
	39	0-23	AT	HPS Elec A Temp (s)	A		See B-II, App B
	39	24, 25,26	TEH	Exp Temps	A		See B-XIX, App B
	40		AT	HPS Elec B Temp(s)	A		
	41		PW	Btry Disch Cur	A		
	42		PW	Btry Volt	A		
	43		PW	Main Bus Volt	A		
	44		PW	Non-Ess Bus Cur	A		
	45		PW	Ess Bus Cur	A		
	46		PW	Solar Array Cur	A		
	47		PW	Shunt Dump Cur	A		
	48		PW	Btry Chg Cur	A		
	49		SD	Xpndr B Sig Str	A		
	50		SD	Xpndr A Static Ph	A		
	51		SD	Xpndr B Static Ph	A		
	52		SD	Xpndr A Rcvr Conv Volt	A		
	53		SD	Xpndr A Xmtr Conv Volt	A		
	54		SD	Xpndr B Rcvr Conv Volt	A		
	55		SD	Xpndr B Xmtr Conv Volt	A		
	56		SD	Xpndr A RF Pwr	A		
	57		SD	Xpndr B RF Pwr	A		
	58		SD	Xpndr A Sig Str	A		
	59		AM	FSS-B Coarse (V01)	A		
122	60		AM	FSS-B Coarse (V02)	A		

Minor Frame Word	Subcom Word	Bit	Unit ID	Telemetry Descriptor	Analog or Digital	Analog Range or Digital States	Remarks
122	61		AM	FSS-B Coarse (VO3)	A		
122	62		AM	FSS-B Fine (VS)	A		
122	63		AM	FSS-B Fine (VC)	A		
123			TL	Encoder Dump	D		
124				Encoder Dump	D		
125				Encoder Dump	D		
126				Sync	D		
127			TL	Sync	D		